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Software Test Plan (STP) for Web Based ML Model Evaluator

Project: Web Based ML Model Evaluator

Version: 1.1

Authors: PES2UG24CS368 – PRATHAM KUMAR - prathamkumar041104@gmail.com
PES2UG24CS324 – NIRUPAMA JAYARAMAN - jnirupama06@gmail.com
PES2UG25CS824 – SHREYAS BELLAD - shreyasbellad2006@gmail.com
PES2UG24CS339 – PARINITHA ACHARYA - parinitha.acharya2006@gmail.com

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1.1	16-09-2026	Project Team	Updated SRS document with revisions and final formatting based on project requirements.	—

Approvals

Role	Name	Signature / Email	Date
Course Coordinator	Sheela Devi M.	sheelam@pes.edu	—

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1. Introduction

1.1 Purpose

This **Software Test Plan (STP)** defines the testing activities for the **Web Based ML Model Evaluator**. The purpose of this document is to describe the testing objectives, scope, test strategy, test environment, responsibilities, schedule, test deliverables, risks, and criteria required to verify that the system satisfies the requirements defined in the Software Requirements Specification (SRS).

The Test Plan provides a common reference for the project team to plan, execute, monitor, and document testing activities. It covers verification of the complete machine-learning evaluation workflow, including user and project management, dataset upload and validation, data preprocessing, model selection, model training, testing, evaluation, comparison, visualization, and report generation. These system functions are defined in the project's SRS.

1.2 Scope

The testing scope includes the major functional and non-functional requirements of the Web Based ML Model Evaluator.

The functional testing will cover user authentication, project management, dataset upload and validation, dataset preview, preprocessing, target-column and task selection, model selection, model training, testing, evaluation metrics, result comparison, visualization, and report generation.

Non-functional testing will cover important quality attributes such as performance, reliability, security, usability/accessibility, data integrity, and maintainability where applicable. Security validation will include areas such as HTTPS, password protection, authorization, file validation, input validation, and security logging, as identified in the SRS.

Testing will focus on the application and its defined interfaces. Internal implementation details of external libraries, frameworks, browsers, operating systems, and third-party services that are outside the project's control are not considered part of the system under test.

1.3 References

The following documents and resources will be used as references during testing:

- **Software Requirements Specification (SRS) – Web Based ML Model Evaluator, Version 1.0**
- **Software Test Plan (STP) – Web Based ML Model Evaluator**
- Requirements Traceability Matrix (RTM) defined in the SRS

- Defined functional and non-functional requirements and their acceptance criteria
- UML use-case diagrams representing the major system workflows
- Test datasets prepared for classification and regression evaluation

The SRS defines the system requirements, acceptance criteria, and traceability information that will be used as the primary basis for designing and executing test cases.

1.4 Definitions and Abbreviations

Term	Definition
STP	Software Test Plan
SRS	Software Requirements Specification
RTM	Requirements Traceability Matrix
ML	Machine Learning
UI	User Interface
API	Application Programming Interface
CSV	Comma-Separated Values
NFR	Non-Functional Requirement
FR	Functional Requirement
UAT	User Acceptance Testing
HTTPS	Hypertext Transfer Protocol Secure
N/A	Not Applicable

1.5 Testing Objective

The primary objective of testing is to verify that the Web Based ML Model Evaluator performs its specified functions correctly and consistently, satisfies the defined non-functional and security requirements, and supports the complete workflow from dataset submission to model evaluation and reporting.

Testing will also ensure that identified requirements are traceable to appropriate test cases through the **Requirements Traceability Matrix (RTM)**. The acceptance criteria defined in the SRS require successful verification of the major dataset-processing, model-training, testing, evaluation, comparison, and reporting workflows.

2. Test Items

The following system modules, features, and interfaces of the **Web Based ML Model Evaluator** will be considered as test items. Testing will verify that each component performs its intended function and interacts correctly with the other components of the system. The test items are derived from the functional workflow and modules defined in the SRS.

2.1 User Authentication and Account Management

This includes user registration, login, logout, session handling, password protection, and access control. Testing will verify that only authorized users can access protected system functions.

2.2 Project Management

This module manages the creation, viewing, updating, and deletion of machine-learning evaluation projects. Testing will verify that project information is correctly stored and associated with the respective user.

2.3 Dataset Upload and Validation

This includes uploading supported datasets, validating file format and structure, checking invalid or corrupted files, and displaying appropriate validation messages.

2.4 Dataset Preview and Basic Analysis

The system shall display uploaded dataset information such as rows, columns, data types, and basic dataset characteristics. Testing will verify that the displayed information corresponds correctly to the uploaded dataset.

2.5 Data Preprocessing

This includes target-column selection, task selection, missing-value handling, categorical encoding, feature scaling where applicable, and train-test data splitting. Testing will verify that preprocessing is correctly applied before model training.

2.6 Model Selection

This module allows users to select supported machine-learning algorithms for classification or regression tasks and configure relevant model parameters.

2.7 Model Training

Testing will verify that the selected model is trained using the prepared training dataset and that the system correctly handles successful, invalid, and failed training operations.

2.8 Model Testing and Evaluation

This includes testing trained models using unseen test data and calculating appropriate evaluation metrics such as **accuracy, precision, recall, F1-score, MAE, RMSE, and R²**, depending on the selected task.

2.9 Model Comparison and Result Visualization

This module presents evaluation results using tables, charts, and comparison views. Testing will verify the correctness of displayed metrics and visual representations.

2.10 Evaluation Result Storage and History

Testing will verify that completed evaluation results are correctly stored and that users can access previously generated results associated with their projects.

2.11 Report Generation and Download

This includes generation of evaluation reports containing relevant dataset, model, metric, and result information. Testing will verify that reports are generated correctly and can be downloaded.

2.12 Error Handling and Status Management

The system shall provide appropriate error messages and processing-status information for operations such as invalid uploads, preprocessing failures, model-training failures, and other unsuccessful operations.

2.13 Security and Access Control

Security-related test items include **HTTPS communication, password protection, authorization, file validation, input validation, and security logging**. These will be tested separately as part of security validation.

2.14 Performance, Reliability, and Accessibility

The system will also be tested against relevant non-functional requirements, including page and dataset-preview response time, system availability, data integrity, accessibility, and maintainability.

Test Items Summary

S. No.	Test Item
1	User Authentication and Account Management
2	Project Management
3	Dataset Upload and Validation
4	Dataset Preview and Basic Analysis
5	Data Preprocessing
6	Model Selection
7	Model Training
8	Model Testing and Evaluation
9	Model Comparison and Result Visualization
10	Evaluation Result Storage and History
11	Report Generation and Download
12	Error Handling and Status Management
13	Security and Access Control
14	Performance, Reliability, and Accessibility

3. Features to be Tested

The following features of the **Web Based ML Model Evaluator** will be tested based on the functional and non-functional requirements defined in the SRS. Testing will verify both individual functions and their interaction across the complete workflow, from user authentication and dataset upload to model evaluation, comparison, and report generation.

3.1 Functional Features

S. No.	Feature to be Tested	SRS Requirement IDs
1	User Registration and Authentication – Verify registration, valid/invalid login, session handling, and logout.	ML-F-001 to ML-F-004
2	Project Management – Verify creation, renaming, deletion, and management of evaluation projects.	ML-F-003
3	Dataset Upload and Validation – Verify supported file upload, file-size/format validation, and rejection of invalid datasets.	ML-F-010 to ML-F-013
4	Dataset Preview and Analysis – Verify display of rows, columns, data types, and basic dataset information.	ML-F-014 to ML-F-017
5	Data Preprocessing – Verify target selection, task selection, missing-value handling, encoding, scaling, and train-test splitting.	ML-F-020 to ML-F-025
6	Model Selection and Configuration – Verify selection of supported classification/regression models and parameter configuration.	ML-F-026 to ML-F-029
7	Model Training – Verify model training, training progress/status, cancellation, model storage, and training-error handling.	ML-F-030 to ML-F-033
8	Model Testing and Evaluation – Verify evaluation using held-out test data and calculation of appropriate classification/regression metrics.	ML-F-040 to ML-F-043
9	Results Comparison and Visualization – Verify side-by-side comparison and graphical presentation of model evaluation results.	ML-F-050 to ML-F-051
10	Preferred Model Selection – Verify that the user can mark a preferred model for a project.	ML-F-052
11	Report Generation – Verify generation and download of evaluation reports containing relevant dataset, model, parameter, and metric information.	ML-F-053
12	Previous Results – Verify retrieval and display of previously stored evaluation results.	ML-F-054

The SRS specifically defines the model-evaluation workflow around held-out test data, classification metrics such as accuracy, precision, recall and F1-score, regression metrics such as MAE, RMSE and R^2 , and result comparison/reporting functions.

3.2 Non-Functional Features

S. No.	Feature to be Tested	SRS Requirement ID
1	Application Performance – Verify that commonly used pages load within the specified response-time target.	ML-NF-001
2	Dataset Preview Performance – Verify dataset preview generation within the specified time for supported datasets up to 10 MB.	ML-NF-002
3	System Reliability – Verify that the system meets the defined availability requirement.	ML-NF-003
4	Error Handling and Usability – Verify that processing failures produce clear messages without exposing internal system details.	ML-NF-004
5	Accessibility – Verify keyboard navigation, descriptive labels, and sufficient colour contrast.	ML-NF-005
6	Data Integrity – Verify consistency of stored project, model-run, and evaluation-result information after completed operations.	ML-NF-006

These six NFRs and their measurable acceptance criteria are explicitly defined in the SRS.

3.3 Security Features

Security-related testing will verify:

- Secure user authentication and password protection.
- Authorization and protection of authenticated sessions.
- HTTPS-secured communication.
- Dataset/file validation.
- Input validation against invalid or unsafe inputs.
- Security-related logging without exposing sensitive information.

These security requirements are part of the SRS and will be validated through dedicated security test cases.

4. Features Not to be Tested

The following features and activities are **outside the current testing scope** of the Web Based ML Model Evaluator. These exclusions help keep testing focused on the requirements and system boundaries defined in the SRS.

4.1 Internal Implementation of Third-Party Libraries

The internal implementation and correctness of external machine-learning libraries, frameworks, and supporting packages will not be tested. Testing will focus only on whether the application correctly uses these components.

4.2 Internal Working of Machine-Learning Algorithms

The mathematical or internal implementation of algorithms such as classification and regression models will not be independently verified. Testing will verify that the system correctly selects, trains, tests, and evaluates the supported models.

4.3 External Services and Infrastructure

The internal operation of external services, cloud infrastructure, internet service providers, operating systems, and browser implementations is outside the project's direct testing scope. The application will only be tested for its interaction with the supported environment.

4.4 Large-Scale Production Deployment

Testing of large-scale production deployment, distributed infrastructure, and enterprise-level load beyond the limits specified for the project is not included in the current scope.

4.5 Unsupported Dataset Formats

Datasets using file formats that are not supported by the application will not be evaluated as valid inputs. Testing will instead verify that unsupported or invalid files are correctly rejected and appropriate error messages are displayed.

4.6 Development and Maintenance Activities Outside the Current Build

Features or capabilities that are not implemented in the current project build and are planned only for future development will not be tested in this test cycle.

4.7 Summary of Exclusions

S. No.	Feature / Activity Not Tested	Reason
1	Internal implementation of third-party libraries	Outside application control
2	Mathematical correctness of ML algorithms	Algorithms are used through supported ML components
3	Internal operation of external infrastructure/services	Outside project boundary
4	Large-scale production deployment	Not part of current project scope
5	Unsupported dataset formats as valid inputs	Only supported formats are within scope
6	Future/unimplemented features	Not available in the current build

The exclusions above are consistent with the SRS approach, which focuses testing on the application's defined functionality, interfaces, security requirements, performance, accessibility, reliability, and data integrity rather than the internal implementation of external components.

5. Test Approach / Strategy

The testing strategy for the **Web Based ML Model Evaluator** will use a combination of functional and non-functional testing techniques. Testing will be performed progressively, starting from individual modules and continuing through their integration and complete system workflow. The approach is based on verifying the requirements and acceptance criteria defined in the SRS.

5.1 Functional Testing

Functional testing will verify whether each system function behaves according to its specified requirement. The following areas will be tested:

- User registration, login, logout, and session handling.
- Project creation and project management.
- Dataset upload, validation, and preview.
- Target-column and task-type selection.
- Data preprocessing and train-test splitting.
- Machine-learning model selection and configuration.
- Model training and training-status handling.
- Model testing using the held-out test dataset.
- Calculation of classification and regression evaluation metrics.
- Model comparison and result visualization.
- Preferred-model selection.
- Evaluation-history access.
- Evaluation report generation and download.
- Error handling for invalid inputs and failed operations.

5.2 Integration Testing

Integration testing will verify whether different modules communicate and work correctly together.

Important integration flows include:

- **Authentication → Project Management**
- **Project Management → Dataset Upload**
- **Dataset Upload → Validation → Preview**
- **Preprocessing → Model Selection**
- **Preprocessing → Model Training**
- **Model Training → Model Testing**
- **Model Testing → Evaluation Metrics**
- **Evaluation → Comparison and Visualization**
- **Evaluation Results → Report Generation**

- **Project/Data → Database Storage**

The objective is to ensure that data is transferred correctly between connected modules and that the output of one component can be correctly consumed by the next component.

5.3 System Testing

System testing will evaluate the complete application as an integrated system.

A complete end-to-end workflow will be tested:

Login



Create Project



Upload Dataset



Validate & Preview Dataset



Preprocess Dataset



Select ML Task & Model



Train Model



Test Model



Evaluate Performance



Compare Results



Visualize Results



Generate Report

The objective is to verify that the complete workflow operates correctly and satisfies the requirements specified in the SRS.

5.4 Performance Testing

Performance testing will verify whether the system satisfies the response-time requirements defined in the SRS.

Testing will include:

- Measuring response time of commonly used pages.
- Measuring dataset-preview generation time for supported datasets.
- Observing system behaviour during model-training operations.
- Checking whether long-running operations provide appropriate processing-status information.

The SRS specifies response-time targets for commonly used pages and dataset preview operations. These targets will be used as the basis for performance validation.

5.5 Usability and Accessibility Testing

Usability testing will verify whether users can understand and operate the application effectively.

Testing will include:

- Checking clarity of labels, buttons, forms, and instructions.
- Verifying understandable error and validation messages.
- Checking consistency of the user interface.
- Testing keyboard navigation.
- Checking descriptive labels for interface controls.
- Checking colour contrast and readability.

These checks correspond to the usability and accessibility requirements defined in the SRS.

5.6 Reliability and Data Integrity Testing

Reliability and data-integrity testing will verify that the system behaves consistently during normal and error conditions.

Testing will include:

- Repeating important operations to verify consistent behaviour.
- Checking recovery from failed processing operations.
- Verifying that completed model runs and evaluation results are stored correctly.
- Checking that project, dataset, model-run, and evaluation information remains consistent.
- Verifying that errors do not corrupt previously stored results.

5.7 Security Validation

Security validation will verify that the application protects user accounts, uploaded datasets, evaluation information, and other application data according to the security requirements defined in the SRS.

5.7.1 Authentication Testing

Testing will verify:

- Registration with valid and invalid information.
- Login using valid credentials.
- Rejection of invalid credentials.
- Logout behaviour.
- Protection of authenticated sessions.
- Password protection and secure password handling.

5.7.2 Authorization Testing

Testing will verify that users can access only the projects, datasets, and evaluation information they are authorized to access.

Test cases will attempt unauthorized access to protected resources and verify that access is correctly denied.

5.7.3 File Upload Security Testing

Dataset-upload functionality will be tested for:

- Unsupported file types.
- Invalid or corrupted files.
- Files exceeding the permitted size.
- Malformed dataset structures.
- Unsafe file names or file contents where applicable.

The system should reject invalid uploads and provide an appropriate error message without exposing internal system information.

5.7.4 Input Validation Testing

User-provided inputs will be tested with:

- Empty values.
- Invalid formats.
- Unexpected characters.
- Incorrect parameter values.
- Boundary values.

The objective is to verify that invalid input is safely handled and does not cause unintended system behaviour.

5.7.5 Data Protection Testing

Testing will verify that:

- User passwords are not stored in plain text.
- Sensitive information is appropriately protected.

- User data and evaluation results are accessible only to authorized users.
- Secure communication is used where required.
- Error messages do not reveal sensitive internal information.

5.7.6 Security Logging Testing

Security-related events such as authentication failures and relevant security events will be checked to ensure that they are recorded appropriately without unnecessarily exposing sensitive information.

The SRS specifically identifies **HTTPS, password protection, authorization, file validation, input validation, and security logging** as security requirements, which will form the basis of these validation activities.

5.8 Overall Testing Strategy Summary

Testing Type	Main Purpose
Functional Testing	Verify individual system functions
Integration Testing	Verify interaction between modules
System Testing	Verify the complete end-to-end workflow
Performance Testing	Verify response-time and processing requirements
Usability Testing	Verify ease and clarity of system usage
Accessibility Testing	Verify accessibility requirements
Reliability Testing	Verify consistent operation and error recovery
Data Integrity Testing	Verify correctness and consistency of stored data
Security Validation	Verify authentication, authorization, input/file security, and data protection

6. Test Environment

The test environment will consist of the hardware, software, development tools, browsers, database, machine-learning libraries, and datasets required to execute the test cases for the Web Based ML Model Evaluator.

6.1 Hardware Environment

- Standard development/testing laptops or desktop computers.
- Minimum **8 GB RAM** recommended for running the web application and ML experiments.
- Adequate storage for application files, datasets, test results, and reports.
- Stable internet/network connection for accessing the web application and required development resources.

6.2 Software Environment

Category	Environment / Tool
Operating System	Windows / Linux
Backend Language	Python
Web Framework	Django
Database	SQLite / PostgreSQL
ML Framework	Scikit-learn
Data Processing	Pandas, NumPy
Frontend	HTML, CSS, JavaScript
UI Framework	Bootstrap
Visualization	Chart.js / Plotly.js
Version Control	Git and GitHub

6.3 Browser Environment

The application will be tested using commonly used modern browsers, including:

- Google Chrome
- Mozilla Firefox
- Microsoft Edge

Browser testing will verify correct page rendering, navigation, forms, file uploads, visualizations, and report downloads.

6.4 Test Datasets

Testing will use representative **tabular datasets** suitable for:

- Classification

- Regression
- Valid dataset-upload testing
- Invalid dataset-upload testing
- Missing-value handling
- Categorical-data processing
- Dataset validation and preview

The datasets will be selected to cover different input conditions and processing scenarios.

6.5 Testing Tools

The following tools may be used during testing:

- **Git/GitHub** – version control and test-related change tracking.
- **Browser Developer Tools** – interface and browser-level debugging.
- **Django testing utilities** – application and backend testing.
- **Python/Scikit-learn utilities** – ML workflow validation.
- **Postman or equivalent API testing tool** – API/interface testing where applicable.
- **Spreadsheet/document tools** – recording test results and preparing test reports.

7. Test Schedule

Testing activities will be planned and completed within the project timeline, with the **final Test Plan and associated testing activities targeted for completion by 15-11-2026.**

Phase	Testing Activity	Planned Timeline
1	Test planning and requirement review	16-09-2026 to 20-09-2026
2	Test scenario and test-case preparation	21-09-2026 to 30-09-2026
3	Test environment and test-data preparation	01-10-2026 to 07-10-2026
4	Functional testing	08-10-2026 to 20-10-2026
5	Integration and system testing	21-10-2026 to 30-10-2026
6	Performance, usability, accessibility and reliability testing	31-10-2026 to 05-11-2026
7	Security validation	06-11-2026 to 09-11-2026
8	Defect correction and regression testing	10-11-2026 to 12-11-2026
9	Final test execution, documentation and review	13-11-2026 to 15-11-2026

8. Test Deliverables

The following documents, records, and results will be produced during the testing process of the **Web Based ML Model Evaluator**.

S. No.	Test Deliverable	Description
1	Test Plan	Defines the overall testing scope, strategy, environment, schedule, responsibilities, risks, and testing activities.
2	Test Scenarios	Identifies the major workflows and situations that need to be tested.
3	Test Cases	Contains detailed test steps, input data, expected results, actual results, and test status.
4	Test Data	Valid, invalid, boundary, and representative datasets used during testing.
5	Requirements Traceability Matrix (RTM)	Maps SRS requirements to corresponding test cases and test results.
6	Defect/Bug Reports	Records identified defects, their severity, status, and corrective actions.
7	Test Execution Report	Summarizes executed test cases, passed/failed cases, and observed results.
8	Performance Test Results	Records response-time and other relevant performance measurements.
9	Security Test Results	Documents authentication, authorization, file-upload, input-validation, and data-protection testing results.
10	Regression Test Results	Confirms that defect fixes and system changes have not adversely affected existing functionality.
11	Final Test Summary Report	Provides the overall testing outcome, major findings, unresolved issues, and final test status.

9. Roles & Responsibilities

Testing responsibilities will be distributed among all four team members to ensure that functional, non-functional, security, documentation, and integration activities are covered

S. No.	Name of Student	SRN	Testing Role	Responsibilities
1	Pratham Kumar	PES2UG24CS368	Test Lead & Integration Testing Coordinator	Prepare and coordinate the overall testing strategy; review SRS requirements for testability; coordinate test-case preparation; handle integration and end-to-end system testing; maintain the RTM; coordinate defect verification and regression testing; consolidate test results and prepare the final test summary.
2	Nirupama Jayaraman	PES2UG24CS324	Functional Testing	Prepare and execute functional test cases for authentication, project management, dataset upload, validation, preview, and preprocessing workflows; record actual results and report functional defects.
3	Shreyas Bellad	PES2UG25CS824	ML Workflow & Performance Testing	Test model selection, model training, model testing, evaluation metrics, comparison, visualization, and report-generation workflows; prepare relevant ML test datasets; perform performance-related testing and record observations.
4	Parinitha Acharya	PES2UG24CS339	Security, Usability & Accessibility Testing	Test authentication and authorization controls, file-upload and input validation, data protection, security logging, usability, accessibility, and error-message behaviour; document security and usability-related defects.

10. Risks & Mitigation

The following risks may affect the testing activities of the Web Based ML Model Evaluator. Appropriate mitigation measures will be used to reduce their impact on the testing schedule and results.

S. No.	Testing Risk	Possible Impact	Mitigation
1	Incomplete or changing requirements	Test cases may become outdated or miss newly introduced requirements.	Review the SRS before test-case preparation and update the RTM whenever requirements change.
2	Insufficient test datasets	Some ML workflows and edge cases may not be adequately tested.	Prepare multiple representative datasets covering classification, regression, missing values, invalid inputs, and boundary conditions.
3	Model training takes longer than expected	Delays in functional and system testing.	Use appropriately sized test datasets and record long-running operations separately during performance testing.
4	Defects discovered late	Reduced time available for correction and regression testing.	Perform testing incrementally and report defects as soon as they are identified.
5	Integration failures between modules	End-to-end workflow may not function correctly even when individual modules work.	Conduct integration testing progressively and verify data flow between connected modules.
6	Browser/environment differences	User-interface or functionality issues may occur in different environments.	Test the application on supported browsers and document environment-specific issues.
7	Security vulnerabilities in file uploads or inputs	Unauthorized or unsafe inputs may affect application security.	Perform dedicated file-upload, input-validation, authentication, authorization, and data-protection testing.
8	Insufficient time before final deadline	Some testing activities may remain incomplete.	Follow the planned schedule and reserve the final days for defect correction, regression testing, and final verification before 15-11-2026 .
9	Test environment instability	Test execution may be interrupted or produce inconsistent results.	Maintain a common test configuration and document environment details before execution.

10	Incomplete test documentation	Difficulty demonstrating requirement coverage and test completion.	Maintain test cases, execution results, defect records, and RTM continuously throughout
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11. Assumptions & Dependencies

Testing of the Web Based ML Model Evaluator depends on the availability of a stable application build, required datasets, software libraries, database services, and testing tools.

S. No.	Assumption / Dependency	Description
1	Stable Application Build	A testable version of the application will be available before formal test execution.
2	Test Datasets	Suitable valid, invalid, and representative datasets will be available for testing ML workflows.
3	Python & ML Libraries	Required libraries such as Pandas, NumPy, and Scikit-learn will be installed and compatible with the application.
4	Database Availability	The configured database will be available for storing and retrieving required application data.
5	Development Environment	Required Python, Django, frontend, and supporting dependencies will be properly configured.
6	Browser Availability	Supported browsers such as Chrome, Firefox, and Edge will be available for UI testing.
7	Test Tools	Required testing, debugging, version-control, and documentation tools will be available.
8	Requirement Stability	The SRS requirements and requirement IDs will remain sufficiently stable during test-case preparation.
9	Team Availability	Team members will be available to execute their assigned testing activities and report results.

12. Suspension & Resumption Criteria

Testing may need to be temporarily suspended when the application or test environment is not suitable for reliable test execution.

12.1 Suspension Criteria

Testing will be suspended when:

- A **critical defect** prevents major application workflows from functioning.
- The application build is unstable or repeatedly crashes.
- Required database services or essential dependencies are unavailable.
- Required test datasets are unavailable or corrupted.
- A major requirement or application workflow is changed and existing test cases become invalid.
- The test environment becomes unavailable or unreliable.
- Test execution results cannot be obtained reliably.

12.2 Resumption Criteria

Testing will resume when:

- Critical blocking defects have been fixed or an approved workaround is available.
- A stable application build is provided.
- Required databases, libraries, and services are operational.
- Required test datasets are restored or replaced.
- Updated requirements and test cases are reviewed where necessary.
- The test environment is restored to a stable condition.

After resumption, **previously affected test cases will be re-executed**, followed by appropriate regression testing.

13. Test Case Management & Traceability

Test cases will be mapped to the requirements defined in the SRS to ensure that every testable requirement is verified. Each requirement will have a unique **Requirement ID**, and each test case will have a unique **Test Case ID**. The basic traceability relationship will be: **SRS Requirement ID → Test Case ID → Test Result → Defect ID (if applicable)**

13.1 Test Case Identification

Test cases will follow a consistent naming convention:

- **TC-AUTH-XXX** – Authentication-related test cases
- **TC-PROJ-XXX** – Project management test cases
- **TC-DATA-XXX** – Dataset-related test cases
- **TC-ML-XXX** – ML model selection/training/testing
- **TC-EVAL-XXX** – Model evaluation and results
- **TC-SEC-XXX** – Security test cases
- **TC-PERF-XXX** – Performance test cases
- **TC-UI-XXX** – UI/usability/accessibility test cases

13.2 Traceability Matrix

The RTM will maintain the relationship between SRS requirements and corresponding test cases.

Requirement ID	Requirement / Feature	Test Case ID(s)	Test Type	Status
FR-XXX	User authentication	TC-AUTH-001, TC-AUTH-002	Functional/Security	Not Executed
FR-XXX	Project creation and management	TC-PROJ-001	Functional	Not Executed
FR-XXX	Dataset upload and validation	TC-DATA-001, TC-DATA-002	Functional/Security	Not Executed
FR-XXX	Model selection and training	TC-ML-001, TC-ML-002	Functional/System	Not Executed
FR-XXX	Model evaluation and results	TC-EVAL-001, TC-EVAL-002	Functional/System	Not Executed
NFR-XXX	Application response time	TC-PERF-001	Performance	Not Executed
NFR-XXX	Data protection	TC-SEC-001	Security	Not Executed

14. Test Metrics & Reporting

Testing metrics will be maintained to measure test progress, requirement coverage, defects, and overall testing status.

14.1 Test Metrics

Metric	Description
Total Test Cases	Total number of test cases planned.
Executed Test Cases	Number of test cases that have been executed.
Passed Test Cases	Number of successfully completed test cases.
Failed Test Cases	Number of test cases that did not produce the expected result.
Blocked Test Cases	Test cases that could not be executed because of a blocking issue.
Test Pass Percentage	Percentage of executed test cases that passed.
Requirement Coverage	Percentage of SRS requirements mapped to test cases.
Defect Count	Total number of defects identified during testing.
Defect Severity	Classification of defects based on their impact, such as Critical, High, Medium, or Low.
Defect Closure Rate	Percentage of reported defects that have been resolved and verified.
Regression Results	Results of re-testing affected functionality after defect fixes.

14.2 Test Reporting

The team will maintain:

- Test Case Execution Report
- Defect/Bug Report
- Requirements Traceability Matrix
- Test Progress Summary
- Performance Test Results
- Security Test Results
- Regression Test Report
- Final Test Summary Report

The final report will summarize the number of tests executed, passed, failed, blocked, identified defects, resolved defects, requirement coverage, and outstanding issues before the final project testing deadline of **15-11-2026**.

15. Approvals

The completed **Software Testing Plan** will be reviewed by each project team member and then submitted to the designated course coordinator for final approval.

Role	Name	Signature / Email	Date
Project Team Member	Pratham Kumar — PES2UG24CS368	prathamkumar041104@gmail.com	16-09-2026
Project Team Member	Nirupama Jayaraman — PES2UG24CS324	jnirupama06@gmail.com	16-09-2026
Project Team Member	Shreyas Bellad — PES2UG25CS824	shreyasbellad2006@gmail.com	16-09-2026
Project Team Member	Parinitha Acharya — PES2UG24CS339	parinitha.acharya2006@gmail.com	16-09-2026
Course Coordinator	Sheela Devi M.	sheelam@pes.edu	—

Project Contribution Details

S. No.	Name of the Student	SRN	Contribution
1	Pratham Kumar	PES2UG24CS368	Tasks 1–5: Introduction, Test Items, Features to be Tested, Features Not to be Tested, and Test Approach / Strategy
2	Parinitha Acharya	PES2UG24CS339	Tasks 6–8: Test Environment, Test Schedule, and Test Deliverables
3	Shreyas Bellad	PES2UG25CS824	Tasks 9–11: Roles & Responsibilities, Risks & Mitigation, and Assumptions & Dependencies
4	Nirupama Jayaraman	PES2UG24CS324	Tasks 12–15: Suspension & Resumption Criteria, Test Case Management & Traceability, Test Metrics & Reporting, and Approvals

THANK_YOU